Gaurav Bhole

+91 9322420299
bholegaurav183@gmail.com
linkedin.com/in/gaurav-bhole18/
gauravbhole.tech
github.com/gauravbhole18

Education

Vellore Institute of Technology **07/2022 – 07/2026**
BTech, Electronics and Communication Engineering (CGPA: 8.50/10)

Relevant Courses

Microcontroller and Microprocessors	Grade: A
Control Systems	Grade: A
Embedded C Programming	Grade: A
Digital Signal Processing	Grade: B

Skills

Programming Languages: C, C++, Embedded C, MATLAB, Java, Python.
Frameworks: FreeRTOS, ROS2, TensorFlow, Pytorch.
Tools and Platforms: Arduino, ESP32, STM32, Raspberry Pi, Linux, Simulink, Cadence
Hardware: Motor drivers , sensors, debugging, Oscilloscope, Function generators.
Soft skills: Presentation, documentation, mentoring
Other Skills: soldering, researching, datasheet reading, circuit design.

Industrial Internships

Intern at Armament Research and Development Establishment, DRDO **05/2025 – 07/2025**

- Performed hardware validation and designed the embedded system architecture for a real-time Fire Control System (FCS) for anti-aircraft applications.
- Wrote drivers for custom made peripherals such as joysticks, encoders, camera, target tracking unit, pan-tilt mechanisms, braking mechanisms etc.
- Developed CMSIS-RTOS v1 firmware with efficient scheduling and safe inter-task communication.
- Implemented safety mechanisms to ensure deterministic system response.
- Made hardware aware design choices for enhanced fault detection, tolerance and rectification.
- Optimised control of high-power devices such as solenoids and brakes with PWM based control algorithm to reduce power consumption and heat dissipation.
- Implemented PID control for autonomous operation of Pan-Tilt-Firing mechanism.
- Tech used: UART, RS422, RS232, RS485, PWM, PID, FreeRTOS, CMSIS-RTOS v1, STM32, STM32cubeIDE.

Drone Development Intern at InsideFPV **04/2023 – 05/2023**

- Designed and built the company’s drone products, integrating hardware and software for optimal flight performance.
- Worked on hardware integration, debugging and drone configuration through programming with tools like Betaflight, INAV, and Mission Planner.

Project Experience

Eye Disease Classification Using Color Fundus Images. **07/2025 – Present**

- Built a multi-class eye disease classification model using CNNs, ViTs, and hybrid architectures on fundus images.
- Improved performance through architecture selection and hyperparameter optimization, surpassing prior benchmark results.

IMU-Based Air Mouse System. **07/2025 – Present**

- Designed an IMU-based air mouse using MPU6050 interfaced with an Arduino/ESP32 over I^2C for real-time motion tracking.
- Implemented sensor fusion (complementary filter) to combine accelerometer and gyroscope data, reducing noise and drift.
- Mapped filtered orientation data to cursor movement and click commands using USB/Bluetooth HID with low-latency performance.

Model Compression Using TensorFlow. **04/2025 – 05/2025**

- Implemented Quantization to achieve 75 Percent reduction in model size and 8 times increase in speed, exploring SIMD to achieve even greater results.
- This can be used to deploy models in resource constrained devices like embedded systems

Flight Controller for Drone **05/2020 – 08/2024**

- Designed and developed a custom drone flight controller using a microcontroller (Arduino / STM32) integrated with a 6-axis IMU (MPU6050) for real-time attitude estimation.
- Implemented sensor fusion and PID-based control loops for roll, pitch, and yaw stabilization, along with PWM-based ESC control for closed-loop motor actuation.
- Supports advanced features including GPS-based position hold and waypoint navigation, autonomous flight modes, altitude hold, failsafe handling, and manual/assisted flight mode switching.

Smart Saver Vehicle – Retrofittable Efficiency Enhancement System **05/2019 – 01/2020**

- Designed and developed a retrofittable, on-demand HHO generation system integrated with ICE vehicles to enhance combustion efficiency through controlled fuel doping, without modifying core engine components.
- Engineered a power-regulated electrolysis cell with safety isolation and fail-safe mechanisms, enabling plug-and-play deployment using the vehicle electrical system.
- Experimentally evaluated mileage and exhaust emissions before and after installation, observing up to ~40% improvement under specific operating conditions, with results validated through certified PUC testing.

Leadership and Extracurricular activities

Coordinator – Clueminati

- Led the end-to-end design and execution of the inaugural edition of Clueminati, a large-scale hybrid technical event.
- Designed the event structure and problem set, and managed logistics, scheduling, and on-ground coordination.
- Established scalable organizational processes that enabled the event to grow into one of the largest technical events in the college in subsequent years.

Senior Core Member - CodeChefVIT

- Mentored electronics and embedded systems projects, guiding students through circuit design, sensor interfacing, and firmware development.
- Served as a judge for hardware and electronics tracks at DEVSOC, evaluating projects on technical depth, implementation quality, and feasibility.

Awards, Honors and Grants

INSPIRE Award Grant Recipient – Department of Science & Technology (DST), Government of India

- Awarded competitive funding for prototype development of the Smart Saver Vehicle project focused on fuel efficiency and emissions reduction.

Online Courses and Certifications

Artificial Intelligence using Google TensorFlow, Optimizing Generative AI on Arm Processors from Edge to Cloud, Foundational MATLAB, MATLAB Onramp, MATLAB Fundamentals.